Data-Driven Software Engineering: Feature Request Apprentice - Managing Software Changes from Feature Requests

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Abstract

The rapid growth of mobile apps has led to increased user-submitted feature requests, often plagued by defects like ambiguity and incompleteness, slowing development and causing misinterpretations.
Our proposed framework uses large language models to automatically detect and resolve defects in feature requests by Interactive dialogues to clarify ambiguous and incomplete feature requests.
Additionally, our framework automates the process of mapping feature requests to the relevant sections of the source code, which is typically a time-consuming task, especially for large projects.

What is the issue?

- **Defects such as ambiguity and incompleteness in natural language**, along with **the complexity of feature localization**, demand significant time and resources. Currently, there is no comprehensive tool to address these challenges efficiently.

Feature Request

Request No: 160

Title: Feature: Group notifications by toot

Description: In the notifications tab, the app tells you that "@realpixelcodeboosted your post" and displays a preview of the toot beneath. Now, when a toot becomes popular and gets quite a few favourites and boosts, it quickly becomes a mess.Instead, my proposal is to group favourite and boost notifications by the respective toot, so the notification would instead look like this:Your post has received 14 boosts from:user1user2user3...[toot preview]

How to resolve?

Defect Detection
Detect natural language defects: ambiguity (lexical, syntactic, semantic, and pragmatic) and incompleteness.

Interactive Refinement
Refine ambiguous or incomplete feature requests through dialogue with the requester.

Source Code Navigation
Automatically map feature requests to the relevant sections of the source code.

Annotation Tool

Request Number - 160 | Request Title - Feature: Group notifications by toot | Request Description - In the notifications tab, the app tells you that "@realpixelcodeboosted your post" and displays a preview of the toot beneath. Now, when a toot becomes popular and gets quite a few favourites and boosts, it quickly becomes a mess.Instead, my proposal is to group favourite and boost notifications by the respective toot, so the notification would instead look like this:Your post has received 14 boosts from:user1user2user3... [toot preview]

Select the category of defect:
(•) Ambiguity ( ) Incompleteness

Select the class of Ambiguity:
( ) Lexical (•) Syntactic ( ) Semantic ( ) Pragmatic ( ) Vagueness

Select the subclass of Ambiguity:
( ) Analytical ( ) Attachment (•) Coordination ( ) Elliptical ( ) N/A ( ) Other

List all the interpretations that are caused by the selected defect (Use semicolons to separate the interpretations).
single grouping for favorite and boost notifications combined; grouping for favorites; grouping for boosts

Does the defect in the selected text affect the downstream software engineering task (e.g., design and implementation)?
(•) Yes ( ) No

What instance(s) of the app design and/or implementation would be affected by this defect? For example, data structure, design size, GUI element design, API usage, etc.
GUI design

Please explain the reasons behind your assessment of the selected text as defect.
It is ambiguous as to whether there should be a single grouping for favorite and boost notifications combined, or a grouping each for favorites and another grouping for boosts. The later example given

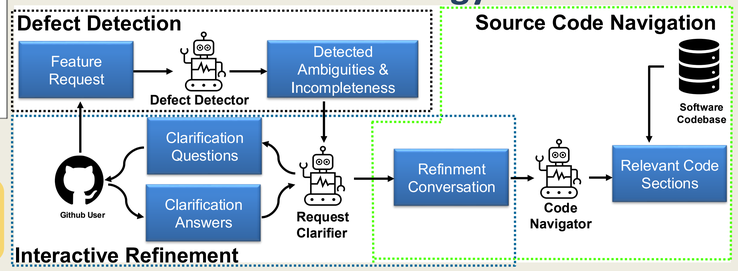
Dataset Detail

APP	Signal	Mastodon
# Samples	414	53

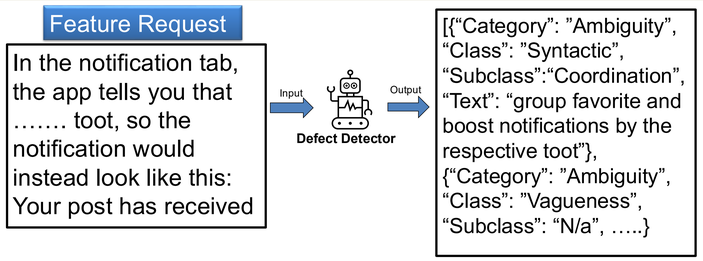
Annotation Format

{"Category":"**Ambiguity**", "Class":"**Syntactic**", "Subclass":"**Coordination**", "Reason":"**It is ambiguous as to whether there should be a single grouping for favorite and boost notifications combined, or a grouping each for favorites and another grouping for boosts. The later example given does not appear to answer this, either, since it does not mention favorites.**", "Questions":"**How should the grouping of favorites and boosts take place? Should there be separate groupings for favorites and boosts per toot?**", "Interpretations":"**single grouping for favorite and boost notifications combined; grouping for favorites; grouping for boosts**", "affectsDownstream":"**Yes**", "affectDescription":"**GUI design**", "text":"**group favourite and boost notifications by the respective toot,**", "start":**357**,"end":**420**}

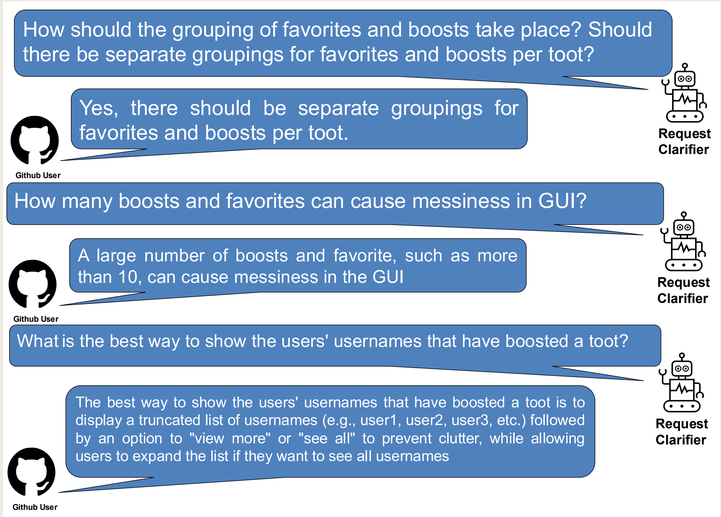
Methodology



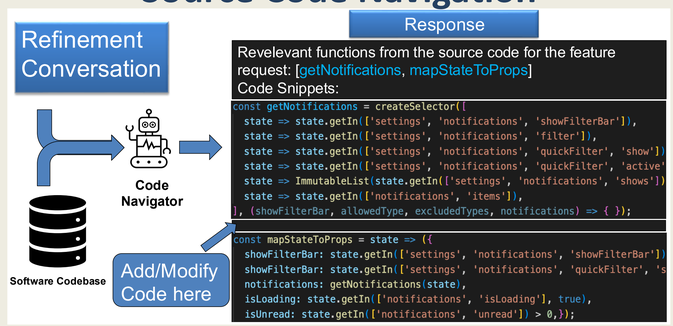
Defect Detection



Interactive Refinement



Source Code Navigation

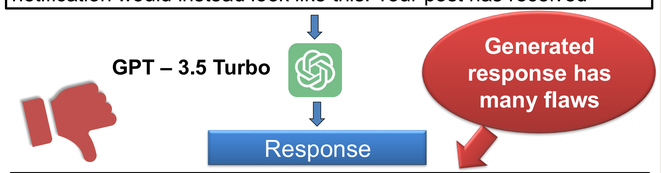


Preliminary Work

Baseline Study: Ambiguity Detection

Prompt: Task + Feature Request

Identify all categories of ambiguity in the following text. The ambiguity categories are lexical, syntactic, semantic, pragmatic, and vagueness.
"In the notification tab, the app tells you that ……. toot, so the notification would instead look like this: Your post has received"



1. Lexical: The use of the term "toot" may be unfamiliar or ambiguous to some readers.
2. Semantic: The phrase "boost notifications" could refer to either notifications of a boosted toot or notifications for boosts given to the user's toot.
3. Syntactic: The phrase "got quite a few favourites and boosts" could be interpreted as the user receiving favorites and boosts, or the toot itself receiving them.
4. Pragmatic: The request description does not specify if this feature would apply to all types of notifications or only certain ones.
5. Vagueness: The phrase "a few" in the request description is subjective and does not give a specific number."

Grounded Analysis of Feature Requests

Feature Request Distribution

	Mastodon	Signal
Feature Requests With Commit	9	87
Feature Requests Without Commit	-	327
Reviewed Requests	8	143
Cleanup Issued	-	184

Closed Feature Requests without Commit ID

	Mastodon	Signal
Unclear or Vague Request	1	1
Added to New App Version	3	37
Duplicate Request	2	40
Referred to Community Forum	2	7
Incomplete Discussion	-	7
Bug Instead of Feature Request	-	4
Referred to Commit Page	-	15
Rejected	-	32

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